

Course Code: CS1280		
Name of the Programme: B.Sc. Computer Science (Hons.)		
Name of the Course: Python Programming for Data Analytics		
Semester: II		
Course credit	No. of hours per week (L + T + P)	Total no. of Teaching hours
3	1+0+4	75
Course Objectives:		
a) Introduce Python syntax, variables, and fundamental programming concepts.		
b) Develop the ability to use Python data structures such as lists, tuples, dictionaries, and sets.		
c) Provide hands-on practice with file handling and data analysis libraries like NumPy and pandas.		
d) Teach data cleaning, preparation, visualization, and basic statistical analysis using Python tools.		

Syllabus:

Module - 1	No. of hours
	03
Python Programming Essentials: Introduction to Python: History, features, applications. Setting up the Python environment: Installing Python and Jupyter Notebook. Basic Python syntax: Variables, data types, and operators, Basic I/O operations, Control Structures: Conditional statements (if, elif, else), loops (for, while, nested loops), and control flow (break, continue, pass). Functions and Modules: Function definition, parameters, return value, recursive functions, and modular programming.	

Module - 2	No. of hours
	03
Data Structures, Class, objects and File Handling: Data Structures: Lists, tuples, sets, and dictionaries, List and dictionary comprehensions, Arrays and 2D arrays: creation, manipulation, iterations, nested loops and applications in grid-based problems (e.g., Sudoku). Classes and objects, Instance variables and methods, File Handling: Reading and writing text files, Working with CSV files. Exception handling: Try and except, user defined exceptions.	

Module - 3	No. of hours
	03
Introduction to NumPy and pandas: NumPy: Creating and working with NumPy arrays (1D, 2D, n-dimensional), Indexing, slicing, and reshaping arrays, Element-wise operations and broadcasting. Pandas: Series	

and DataFrame creation, Basic operations on DataFrames: indexing, slicing, filtering, Handling missing data: dropping and filling missing values. Descriptive Statistics and Inference: central tendencies: mean, median, and mode. Variability and dispersion: range, variance, standard deviation. Analysing data distributions (e.g., skew, kurtosis)

Module - 4	No. of hours
	03
Exploratory data analysis (EDA): Data Cleaning and Preprocessing: Handling missing and duplicated data, Outlier Detection: Q1, Q3 using IQR. Encoding categorical variables (label encoding, one-hot encoding), Scaling and normalization (min-max scaling, z-score normalization). Advanced Operations with Pandas: concatenating, merging, and joining DataFrames, groupBy operations, and aggregation.	

Module - 5	No. of hours
	03
Data Visualization and Analysis with Python: From Basics to Case Studies Matplotlib: line plots, bar charts, histograms, scatter plots, customizing plots (titles, labels, and legends), subplots, figure layout, and visualizing IQR with box plots. Seaborn: Statistical plots: box plots, violin plots, pair plots, correlation heatmaps, customizing Seaborn plots (themes, color palettes). Introduction to Plotly. Case Studies: Exploratory data analysis (EDA) on real-world datasets, Examples from domains such as finance, healthcare, and sports.	

Course outcomes
a) Apply Python programming fundamentals and data structures to build basic programs, handle files, and organize data.
b) Implement data analysis using NumPy and Pandas, including descriptive statistics and data preprocessing techniques.
c) Create data visualizations using Matplotlib and Seaborn to represent data insights effectively.
d) Develop and present a complete data analysis project demonstrating skills in programming, data manipulation, and visualization.

Text Books	
1	McKinney, W. (2017). <i>Python for data analysis</i> (2nd ed.). O'Reilly Media. ISBN: 978-1491957660
2	Zelle, J. M. (2010). <i>Python programming: An introduction to computer science</i> (2nd ed.). Franklin, Beedle & Associates. ISBN: 978-1590282755

Reference books	
1	Sweigart, A. (2015). <i>Automate the boring stuff with Python: Practical programming for total beginners</i> . No Starch Press. ISBN: 978-1593279929

2	VanderPlas, J. (2016). <i>Python data science handbook: Essential tools for working with data</i> . O'Reilly Media. ISBN: 978-1491912058
3	Muller, A. C., & Guido, S. (2016). <i>Introduction to machine learning with Python: A guide for data scientists</i> . O'Reilly Media. ISBN: 978-1449369415

Reference Links	
1	https://www.geeksforgeeks.org/data-analysis-with-python/
2	https://www.freecodecamp.org/learn/data-analysis-with-python
3	https://onlinecourses.nptel.ac.in/noc21_cs45/preview

Lab Programs		[60 Hrs]
1	Setup and Environment: A. Installation and configuration of Anaconda (install file in USB pendrive), Python 3.12 and Jupyter Notebook 7.1.2 in the local environment Getting comfortable with Google Colab environment	
2	Basic Programming and Syntax: A. Simple calculator B. Temperature converter (Celsius to Fahrenheit and vice versa) C. Program to find the largest of three numbers	
3	Functions and Recursion: A. Factorial of a number using recursion B. Word count program	
4	Data Structures: A. Program to perform various operations on lists (insert, delete, find, etc.) B. Program to merge two dictionaries	
5	File Handling: A. Program to read a CSV file and perform basic analysis Getting awareness about local and Google drive folder and file access, renaming, copying, and moving files	
6	NumPy Basics: A. Creating and manipulating NumPy arrays B. Program to perform matrix operations using NumPy	
7	Pandas Basics: A. Creating a DataFrame from scratch and from a CSV file B. Program to clean data (handling missing values, duplicates)	
8	Descriptive Statistics: Use mean(), median(), mode(), min(), max(), var(), std() from Pandas to measure central tendencies and dispersion of data	

9	Data Distribution: Use describe(), quantile(), skew(), kurt(), value_counts() from Pandas for shape and distribution data
10	Categorical Encoding: A. Perform categorical encoding—OneHotEncoder, LabelEncoder from Python
11	Data Scaling: Scale the data using Python: StandardScaler(), Normalizer(), and MinMaxScaler() from sklearn.preprocessing
12	Outlier Detection: A. Visualize outliers using box plot from Matplotlib and Seaborn B. Identify outliers in datasets using Pandas DataFrame and Interquartile Range (IQR)
13	Advanced Pandas Operations: A. Advanced Pandas operations, GroupBy, merging, joining, concatenating DataFrames B. Time series analysis of stock prices (Optional case study)
14	Advanced Visualization with Seaborn: A. Using Seaborn to visualize data distributions B. Creating a pair plot for a DataFrame
15	Correlation: A. Visualize correlation matrix using heatmap from Seaborn to enhance readability
16	Identifying Trends and Relationships: A. Identify the trend of any variable using line chart from Matplotlib and Seaborn B. Visualize the relationship between two correlated columns using scatter plot
17	Category and Numeric Relationships: Visualize the relationship between categorical and a numerical column using bar plot